

Fatemeh S. Ahmadi

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RESEARCH INTERESTS

Biophysics, Cancer Research, Fluid Dynamics, Computational Physics

EDUCATION

M.Sc. in Condensed Matter Physics (1st Rank in Class), KNToosi University of Technology, Tehran, Iran Oct. 2021 – Feb. 2024

GPA: 17.6/20 (3.8/4.0) - *Summa Cum Laude* Equivalent

Thesis title: *Investigation of particle geometry and inhomogeneity on the dynamics in microfluidic domains*

Thesis grade: 20/20

Supervisor: [Prof. H. Hamzeshpour](#), Advisor: [Prof. R. Shaebani](#)

Key achievements:

- **Technical Proficiency:** Developed advanced expertise in computational modelling using COMSOL Multiphysics (Fluid-Structure Interaction module) and data analysis with Python (pandas, NumPy, Matplotlib).
- **Analytical Thinking:** Investigated how particle geometry dictates transport dynamics in confined biological flows using Stokes flow with Poiseuille velocity profiles.
- **Innovation:** Established a predictive framework for passive particle segregation based on geometric parameters.
- **Resilience & Problem-Solving:** Optimised computational parameters to resolve persistent simulation crashes, ensuring project completion under challenging constraints.

B.Sc. in Physics, Shahed University, Tehran, Iran

Sep. 2016 – Jul. 2021

GPA: 15.39/20 (3.1/4.0)

PUBLICATIONS

Ahmadi, F.S., Hamzeshpour, H. & Shaebani, R. (2025). Geometry-Driven Segregation in Periodically Textured Microfluidic Channels. *Under peer review in Physical Review Fluids*. [arXiv:2504.05834](https://arxiv.org/abs/2504.05834)

RESEARCH EXPERIENCE

Mentor

Aug. 2024 - Jan. 2025

- **Title:** *Numerical study of particle deformation with different Young's modulus during passage through microchannels*.
- **Mentoring:** Advised a Physics M.Sc. student on thesis research and methodology.
- **Technical Instruction:** Diagnosed simulation issues and instructed on COMSOL's FSI module, accelerating thesis completion.

Seminar course (Grade 20/20)

May 2022

- **Title:** *Particle shape effect on its motion in a microfluidic channel*
- **Research Synthesis:** Conducted a literature review to map key trends in microfluidic applications for cancer research and proposed future directions.
- **Presentation:** Presented findings to an audience of 60.

Research Assistant

May 2021

- **Title:** *Medical Applications of Laser*
- **Collaboration & Teamwork:** Collaborated with a Physics PhD candidate to conduct a comprehensive literature review.
- **Time Management:** Classified findings to meet critical deadlines under time pressure.

SKILLS

Simulation Software: COMSOL Multiphysics, OpenFOAM (Basic), AutoCAD (Basic)

Programming Languages: Python, C++, Fortran (Basic), Matlab (Basic), Web Design (Basic)

Typesetting: LaTeX, Microsoft Office

Operating Systems: WIN, Linux

Research Software: Mendeley, Zotero, Notion, Obsidian, Inkscape

Languages: English (Fluent), Arabic (Basic), Persian(Native)

COURSES

Theory of Living Matter - Seminar Series (TLMCambridge)	Since 2022
Machine Learning Scientist in Python - DataCamp	Jan. 2025
Data Analyst in Python - DataCamp	Dec. 2024
Molecular Foundations of Medicine - StanfordOnline (EdX)	Nov. 2024
Improve Your English Communication Skills -Georgia Institute of Technology (Coursera)	Oct. 2024
English Grammar and Style -The University of Queensland (EdX)	Sep. 2024

AWARDS

Graduated with First Rank in M.Sc. Program - KNToosi University of Technology	Feb. 2024
Full tuition scholarship - M.Sc. - National Entrance Exam Ranking	Oct. 2021
Full tuition scholarship - B.Sc. - National Entrance Exam Ranking	Sep. 2016

VOLUNTARY WORK

Elderly Care, Local Community, Tehran, Iran	2024 - present
• Empathy & Compassion: Provided dedicated companionship and daily living assistance, demonstrating patience and interpersonal sensitivity.	
Teaching Assistant, KNToosi University of Technology, Tehran, Iran	2022
• Knowledge Transfer: Initiated problem-solving workshops for graduate courses "Advanced Surface Physics" and "Advanced Electrodynamics". • Communication: Generated detailed solution sets and visual aids to enhance student comprehension by clarifying complex concepts.	
Accountant, Local Preschool, Tehran, Iran	2021 - 2022
• Accuracy: Arranged all financial tracking and reporting, ensuring data accuracy and clear communication of budgetary performance.	
Founder & Member of Physics Scientific Association, Shahed University, Tehran, Iran	2017 - 2018
• Leadership & Initiative: Co-founded and managed a student-led organisation aimed at promoting scientific engagement beyond the standard curriculum. • Organisation & Project Management: Organised seminars and academic discussions, coordinating events and participant engagement.	
Physics Tutor, Local Community, Tehran, Iran	2016 - 2017
• Adaptive Communication & Teaching: Designed and delivered personalised physics lessons to local students, focusing on strengthening foundational knowledge and problem-solving skills.	

EXTRA-CURRICULAR ACTIVITIES

- Actively study biophysics and computational biology through specialised courses and seminars.
- Continuously explore emerging technologies and programming languages for their utility in both research and daily life.
- Deconstructing complex narratives in mystery and science fiction.
- Honing problem-solving through strategic games and finding mental clarity through cycling and hiking.
- Exploring diverse cultures through travel and language study.
- Dedicated to the joyful care of my pet.

REFEREES

Prof. Hossein Hamzhepour, KNToosi University of Technology, Tehran, Iran.
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Prof. Reza Shaebani, Saarland University, Saarbrücken, Germany.
Email: shaebani@lusi.uni-sb.de Tel: +49 681 302 3964